

NICHOLAS HO

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EDUCATION

University of Southern California	Los Angeles, CA
Master of Science, Applied Data Science	January 2026–Present (Expected May 2027)
Bachelor of Science, Quantitative Biology	August 2021–December 2025

EXPERIENCE

Walmart Global Tech	Sunnyvale, CA
Data Science Intern – Applied AI, Last Mile Delivery	June 2026–Present

- Develop an anomaly-detection system for Walmart’s Spark delivery platform to surface unusual driver behavior and prioritize potentially fraudulent cases for human review
- Extract and transform 180 days of driver-trip data from BigQuery using SQL, building driver-day behavioral profiles across the five largest Spark Zones
- Evaluated 50+ behavioral features using within-driver variability, between-driver variance, and one-way ANOVA, selecting 20+ signals for personalized anomaly detection
- Built and compared a statistical z-score framework and PCA-based Isolation Forest model, and applied prompt-engineering techniques to generate feature-level explanations and reviewer-friendly investigation rationales

USC Laboratory for Machine Learning, Health, and Biomedicine	Los Angeles, CA
AI Research Intern – Professor Ruishan Liu	May 2025–August 2025

- Wrote Python scripts to preprocess and standardize genomic sequences across formats (FASTA, BED, GZ), enabling uniform integration into a multi-task learning pipeline
- Authored Python tools to convert genomic coordinates from GRCh37 to GRCh38, ensuring compatibility across datasets and improving data alignments for downstream model training
- Conducted literature review and constructed a metadata schema to organize TCGA and other public datasets by predictive task categories, cutting down data retrieval time for model development by 30%

H&R Block	Kansas City, MO
Software Engineering Intern – Virtual Tax Submission Team	May 2024–August 2024

- Developed and deployed a digital consent collection tool leveraging Angular and C#, enhancing user experience for more than 1 million users by decreasing page reload friction
- Achieved 95% unit test coverage and supported compliant software delivery through Azure DevOps pipelines and CI/CD best practices
- Collaborated with product and legal teams to enhance data submission workflows, streamlining consent processes and mitigating user drop-off by ~20%

PROJECTS

NBA Player Stats & Fantasy Prediction Project	December 2025–Present
Python React Flask scikit-learn NumPy SQL	

- Built and trained an NBA stat prediction model in Python leveraging scikit-learn and NumPy, forecasting 9 player performance metrics based on historical data and engineered features (e.g., age trends, minutes projections)
- Developed a dynamic JavaScript (React) frontend with custom HTML/CSS styling to visualize and rank players by predicted stats
- Built a Flask API to serve model predictions and deployed an interactive web application

Healthcare Pricing Prediction	March 2025–December 2025
Python matplotlib Tableau Jupyter Notebook	

- Built an end-to-end regression pipeline in Python utilizing scikit-learn and XGBoost to predict hospital treatment costs, achieving an R^2 above 0.85
- Cleaned and preprocessed 50,000+ healthcare records in Jupyter Notebook by imputing missing values, encoding categorical variables, and scaling features, lowering model training time by 25% and boosting accuracy
- Assessed model accuracy using MAE, RMSE, and R^2 , and visualized key pricing patterns with matplotlib and Tableau dashboards for stakeholder presentation

TECHNICAL SKILLS

Languages: Python, SQL, R, C++, C#, HTML/CSS, JavaScript
Machine Learning & AI: scikit-learn, XGBoost, Isolation Forest, PCA, prompt engineering, feature engineering, model evaluation
Data & Cloud: pandas, NumPy, BigQuery, Google Cloud Platform
Frameworks & Visualization: React, Angular, Flask, Tableau, matplotlib, seaborn
Developer Tools: Azure DevOps, Git, GitHub, Jira